

Q 1. Calculate the minimum uncertainty in the location of (i) a 500 mg mass moving with a speed of 1.5 ms^{-1} , and (ii) an electron (mass = $9.11 \times 10^{-25} \text{ mg}$) moving with a speed of $2.2 \times 10^6 \text{ ms}^{-1}$. In each case the uncertainty in the momentum measurement is $\Delta p = 10^{-3}p$.

Q 2. Fill up the table, considering principal quantum number (n) = 1, 2, 3, and 4.

n	l	m_l	Orbital type	Total no. of orbital(s)	Maximum no of electrons
1					
2					
3					
4					